

Two players take turns taking counters from a pile. On each turn, a player takes 1 counter or 2 counters. Whoever takes the last counter wins.

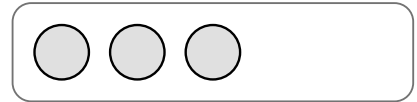
Problem 1: It is your turn. Circle the counters you take now to be sure to win, or cross out the pile if you cannot be sure to win.



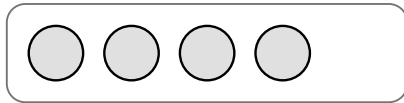
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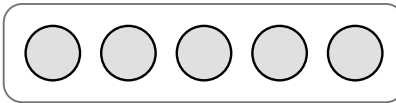
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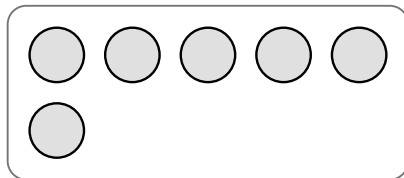


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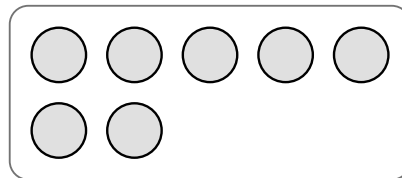
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Problem 2: Circle 1st or 2nd to show which player can be sure to win with each pile.



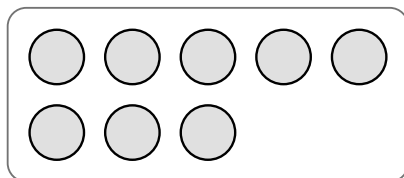
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1st 2nd



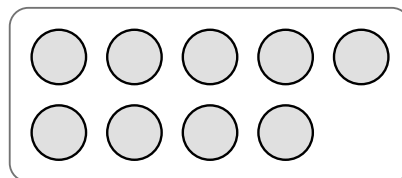
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1st 2nd



8

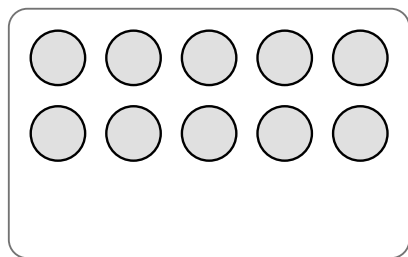
1st 2nd



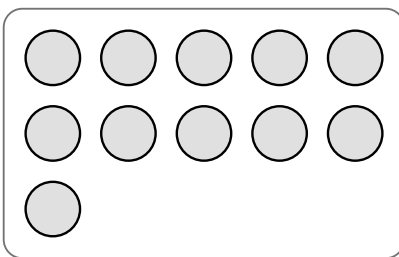
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1st 2nd

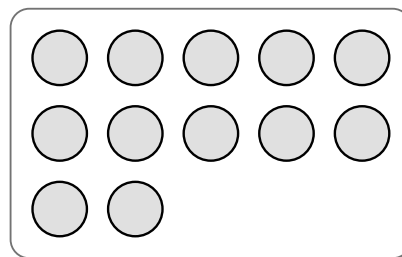
Problem 3: It is your turn with a bigger pile. Circle the counters you take now to be sure to win, or cross out the pile if you cannot be sure to win.



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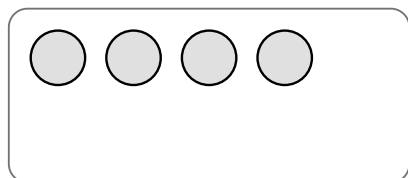
12

Problem 4: A pile is a trap if you cannot be sure to win when it is your turn, like the pile of 3. Put an X on every trap from 1 to 20.

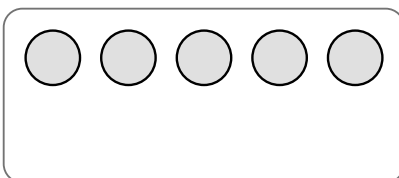
1	2	3	4	5	6	7	8	9	10
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11	12	13	14	15	16	17	18	19	20
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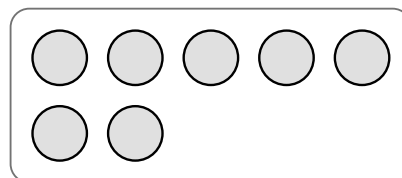
Problem 5: Ben goes first and takes 2 counters. Put a check by each pile where Ben can still be sure to win.



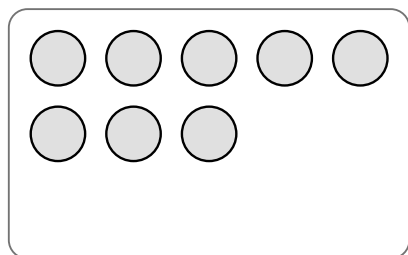
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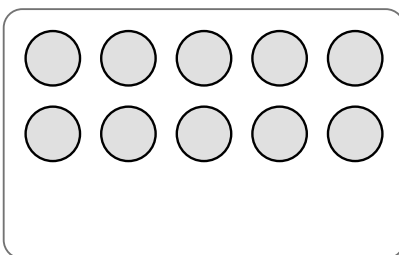
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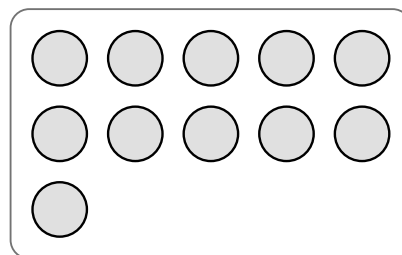
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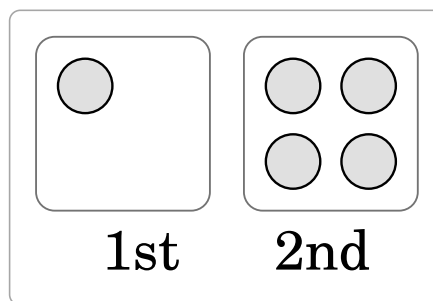
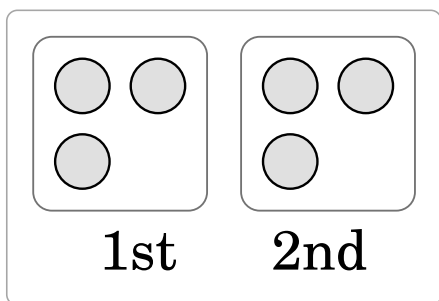
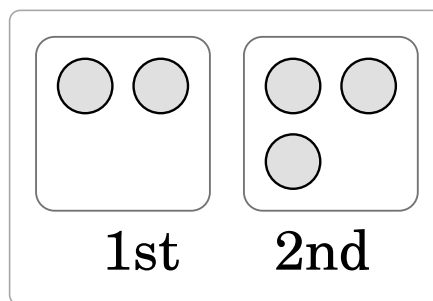
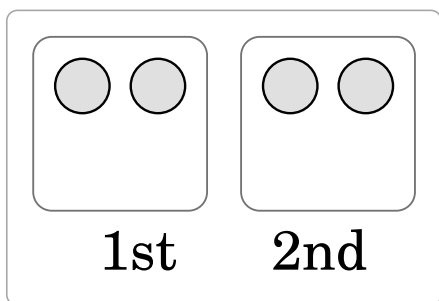
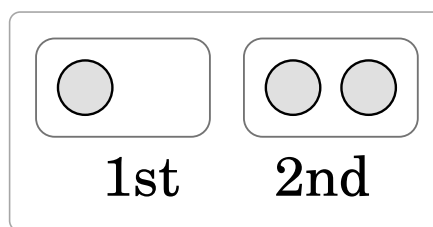
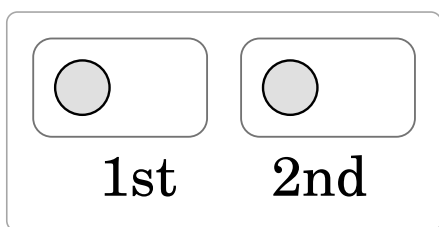
Problem 6: Show every different way a game with 4 counters can go, and then every way a game with 5 counters can go. Taking 1 then 2 is different from taking 2 then 1. Circle the counters taken on each turn. Some rows may stay empty.

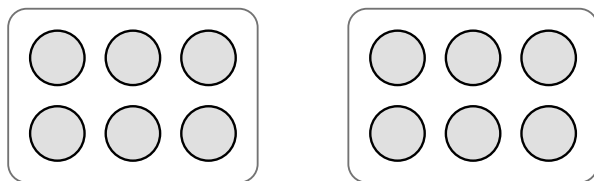
Problem 7: Now each player may take 1, 2, or 3 counters on a turn. Put an X on every trap from 1 to 20.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Problem 8: Now there are two piles, and on your turn you take 1 or 2 counters from one pile. Circle 1st or 2nd to show which player can be sure to win.



Problem 9: Play the game from Problem 8 with two piles of 6 counters. If you go second, how can you win every time, whatever the other player does?



Problem 10: Now there is one pile again. Each player takes 1 or 2 counters on a turn, and whoever takes the last counter loses. Put an X on every trap from 1 to 20.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20